

$$Kn = \frac{\lambda}{L}$$

$$\tau = \mu \frac{du}{dy}$$

$$\tau = m \left| \frac{du}{dy} \right|^{n-1} \frac{du}{dy}$$

$$\mu = \mu_0 \left(\frac{1}{1 + \alpha t + \beta t^2} \right)$$

$$\mu = \mu_0 + \alpha t - \beta t^2$$

$$h = \frac{\sigma \times \pi d \times \cos \theta}{\frac{\pi}{4} d^2 \times \rho \times g} = \frac{4 \sigma \cos \theta}{\rho \times g \times d}$$

$$h^* = \frac{I_G \sin^2 \theta}{A \bar{h}} + \bar{h}$$

$$GM = \frac{I}{\nabla} - BG$$

$$T = 2\pi \sqrt{\frac{K^2}{GM \cdot g}} \quad I_{Y-Y} = \frac{W}{g} K^2$$

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0$$

$$\vec{\omega} = \frac{1}{2}(\nabla \times \vec{V})$$

$$\Gamma = \int_C V \cos \alpha \, ds$$

$$\zeta = \frac{\text{Circulation}}{\text{Area}}$$

$$= \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$

$$\dot{\gamma}_{xy} = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}$$

$$\omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right)$$

$$\omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right)$$

$$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$

$$u = -\frac{\partial \phi}{\partial x}$$

$$v = -\frac{\partial \phi}{\partial y}$$

$$w = -\frac{\partial \phi}{\partial z}$$

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y} + w \frac{\partial}{\partial z}$$

$$1 \text{ Bar} = 100000 \text{ N/m}^2$$

$$1 \text{ Bar} = 0.986923 \text{ atm}$$

$$1 \text{ atm} = 101325 \text{ Pa}$$

Discharge over trapezoidal notch/weir

$$Q = \frac{2}{3} C_{d_1} L \sqrt{2g} \times H^{3/2} + \frac{8}{15} C_{d_2} \times \tan \theta/2 \times \sqrt{2g} \times H^{5/2}$$

Discharge over broad crested weir

$$Q = C_d L h \sqrt{2g(H-h)}$$

$0.4 \leq H/B \leq 1.6 \rightarrow$ Narrow-crested weir

$0.1 \leq H/B \leq 0.4 \rightarrow$ Broad-crested weir

Max discharge over broad crested weir

$$Q = 1.70 C_d L H_1^{3/2}$$

$$H_1 = (H + h_a) = \left[H + \left(\frac{V_a^2}{2g} \right) \right]$$

Hagen Poiseuille equation

$$h_f = \frac{32 \mu \bar{u} L}{\rho g D^2}$$

Time taken to empty reservoir with rectangular notch/weir

$$T = \frac{3A}{C_d \times L \times \sqrt{2g}} \left[\frac{1}{\sqrt{H_2}} - \frac{1}{\sqrt{H_1}} \right]$$

Time taken to empty reservoir with triangular notch/weir

$$T = \frac{5A}{4 \times C_d \times \tan \frac{\theta}{2} \times \sqrt{2g}} \left[\frac{1}{H_2^{3/2}} - \frac{1}{H_1^{3/2}} \right]$$

Bazin's formula

$$Q = m_1 \sqrt{2g} L H_1^{3/2}$$

$$m_1 = \left(0.405 + \frac{0.003}{H_1} \right) \quad H_1 = \left(H + \alpha \frac{V_a^2}{2g} \right)$$

Francis's formula

$$Q = 1.84 (L - 0.1 n H_1) (H_1^{3/2} - h_a^{3/2})$$

- Head of liquid $> 5 \times$ depth of orifice $\rightarrow$ small orifice
- Head of liquid $< 5 \times$ depth of orifice $\rightarrow$ large orifice

Viscous flow through pipes

$$\tau = -\frac{\partial p}{\partial x} \frac{r}{2} \quad u = \frac{1}{4\mu} \frac{\partial p}{\partial x} r^2 - \frac{1}{4\mu} \frac{\partial p}{\partial x} R^2$$

$$= -\frac{1}{4\mu} \frac{\partial p}{\partial x} [R^2 - r^2]$$

Viscous flow through parallel plates

$$\tau = -\frac{1}{2} \frac{\partial p}{\partial x} [t - 2y] \quad u = -\frac{1}{2\mu} \frac{\partial p}{\partial x} [ty - y^2]$$

$$h_f = \frac{p_1 - p_2}{\rho g} = \frac{12\mu \bar{u} L}{\rho g t^2}$$

Time of emptying cylindrical tank through orifice at bottom

$$T = \frac{2A [\sqrt{H_1} - \sqrt{H_2}]}{C_d \cdot a \cdot \sqrt{2g}}$$

Discharge through partially submerged orifice

$$Q_1 = C_d \times b \times (H_2 - H) \times \sqrt{2gH}$$

$$Q_2 = \frac{2}{3} C_d \times b \times \sqrt{2g} [H^{3/2} - H_1^{3/2}]$$

Time of emptying hemispherical tank through orifice at bottom

$$T = \frac{\pi}{C_d \times a \times \sqrt{2g}} \left[\frac{4}{3} R (H_1^{3/2} - H_2^{3/2}) - \frac{2}{5} (H_1^{5/2} - H_2^{5/2}) \right]$$

Darcy Weisbach equation

$$h_f = \frac{4 \cdot f' \cdot LV^2}{2g \cdot d} \quad \text{or} \quad h_f = \frac{f \cdot L \cdot V^2}{d \times 2g}$$

Time of emptying horizontal cylindrical tank through orifice

$$T = \frac{4L}{3C_d \times a \times \sqrt{2g}} \left[(2R - H_2)^{3/2} - (2R - H_1)^{3/2} \right]$$

$$Q = 0.855 \times \text{Area of Mouthpiece} \times \sqrt{2gH}$$

$$Q = 0.5 \times \text{Area of Mouthpiece} \times \sqrt{2gH}$$

$$Q = 0.707 \times \text{Area of Mouthpiece} \times \sqrt{2gH}$$

Submerged/drowned weir

$$Q_1 = \frac{2}{3} C_{d1} L \sqrt{2g} (H_1 - H_2)^{3/2}$$

$$Q_2 = C_{d2} (L \times H_2) \sqrt{2g(H_1 - H_2)}$$

$$Re = \frac{\rho V D}{\mu}$$

Orifice meter

$$Q = C_d A_2 \frac{\sqrt{2g\Delta H}}{\sqrt{1 - C_c^2 (D_2/D_1)^4}} \quad \Delta H = \frac{(p_1 - p_2)}{\gamma}$$

$$Q = C_c A_0 C_v \sqrt{\frac{2g(\rho_m/\rho - 1)}{(1 - C_c^2 A_0^2/A_1^2)} \Delta h}$$

Venturi meter

$$Q = C_d \frac{a_1 a_2}{\sqrt{a_1^2 - a_2^2}} \times \sqrt{2gh}$$

Minor losses

$$h_e = \frac{(V_1 - V_2)^2}{2g} \quad h_c = \frac{V_2^2}{2g} \left[\frac{1}{C_c} - 1 \right]^2$$